Auditing Selection Purity and Transfer Utility under Joint Model and Data Heterogeneity

Anonymous submission

Abstract

A canonical probability interface makes cross-architecture exchange well-defined across language models with different parameters and tokenizers, but cleaner selected client data need not yield more useful server supervision. We introduce a four-arm audit that separates clean-transfer utility, contamination sensitivity, and the complete routing intervention, together with matched-size comparisons that probe selection order separately from subset size. A reused-test TREC decomposition finds that routing raises audited clean fraction from 40.0% to 64.6% but reduces severe-uniform accuracy by $1.42\,pp$ (Holm $p = .0074$). A protocol-frozen role reversal on the 25,000-review IMDb test, with no retained record of prior model assessment on that split, reverses that direction: routing raises clean fraction from 40.2% to 66.5% and improves severe-uniform accuracy by $0.29\,pp$ (95% CI $[0.07, 0.51]$, paired $p = .0149$), so the pre-specified negative-direction gate fails. A post-primary AG News resampling also gives a positive routed-minus-severe effect ($+1.46\,pp$) while remaining below anchor-only by $0.58\,pp$. At matched $K = 128$, routed selection exceeds target-eligible random selection by $0.32\,pp$ (Holm $p = .0020$). The audit-only maximum-attainable-purity arm has a $+0.05\,pp$ point difference from routing (95% CI $[-0.10, +0.19]$; Holm $p = .487$), so these data do not establish an additional gain from higher purity. The opposite TREC and IMDb routing signs, together with AG's split between positive D–C and negative D–A, show that transfer utility is setting- and estimand-specific.

Introduction

Federated fine-tuning promises to adapt language models without moving private client text. Two assumptions make this problem look easier than it is. First, many methods assume compatible client parameters even though deployed models can differ in width, tokenizer, and architecture. Second, evaluations often assume that a participating client's data is uniformly useful for the server task. Real client corpora can instead contain a mixture of useful examples, unrelated tasks, and responses that directly conflict with the target capability.

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These axes interact. Parameter-space averaging may be undefined across model families, but moving to logits or probabilities only solves the interface problem; it does not establish whether transferred signals optimize the target server metric. Conversely, a local data filter that works for one model family may exploit architecture-specific loss scales and fail on another. We ask: *when client architectures differ, does raising audited clean fraction improve the utility of probabilities transferred to the server?*

We make three contributions. First, we operationalize a four-arm audit for heterogeneous client-to-server distillation: public-anchor only (A), clean uniform transfer (B), severely contaminated uniform transfer (C), and severely contaminated routed transfer (D). The adjacent contrasts B–A, C–B, and D–C separate opposing transfer effects hidden by a near-zero endpoint. Second, we separate discovery from protocol-frozen evaluation: a reused-test TREC decomposition generates the hypothesis, an IMDb test with no retained record of prior model evaluation supplies the cross-setting stress test, and its positive D–C effect contradicts the TREC-derived negative direction; AG News supplies prior task-specific evidence. Third, fixed-size random, routed, and maximum-attainable-purity comparisons show useful routed ordering without a detectable gain from still higher purity. Content-addressed protocols and complete run records preserve these inferential roles. This is an evaluation contribution, not a new optimizer, privacy mechanism, or claim that purity and utility are universally unrelated. Figure 1 summarizes the estimands and inferential roles.

Related Work

Federated PEFT and model heterogeneity. FedAvg aggregates compatible client parameters (McMahan et al. 2017), while FedMD and FedDF use public-data distillation to relax the shared-architecture assumption (Li and Wang 2019; Lin et al. 2020). LoRA makes adaptation economical by training low-rank updates (Hu et al. 2022), but factor exchange does not by itself resolve architectural incompatibility. FedBiOT compresses a server LLM for client-side adaptation (Wu et al. 2024). HetLoRA supports heterogeneous client LoRA ranks (Cho et al. 2024), while FLoRA avoids cross terms through stacking (Wang et al. 2024). FedMKT transfers knowledge between server LLMs and client SLMs (Fan et al. 2025). These works establish that

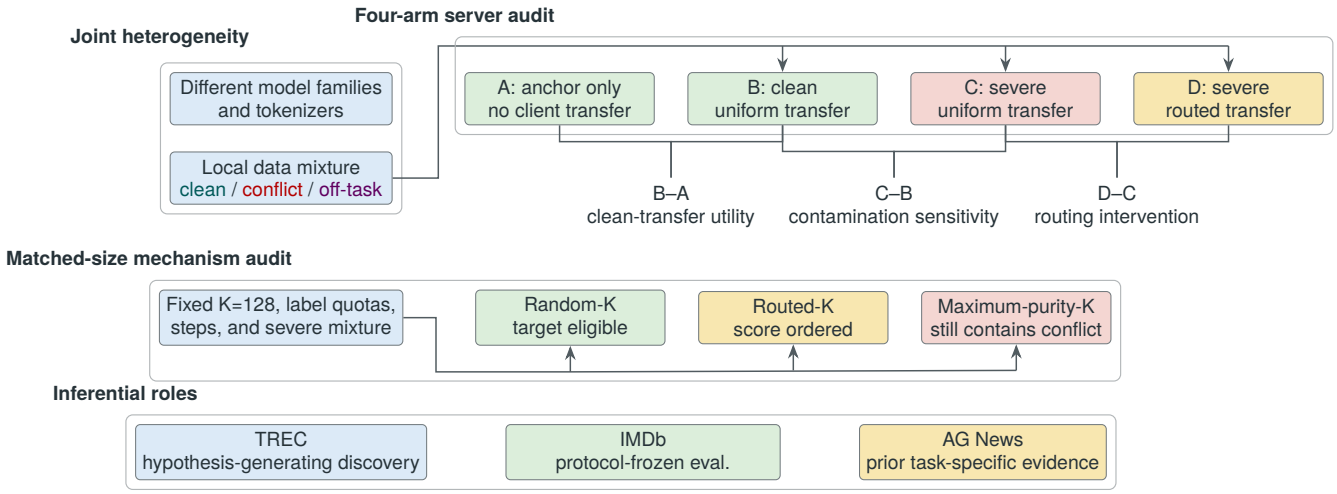


Figure 1: Evidence-aware four-arm audit. A–D separate clean-transfer utility, contamination sensitivity, and the complete routing intervention. The matched-size audit holds K, claimed-label quotas, optimizer steps, and the severe mixture fixed. TREC is hypothesis-generating discovery, IMDb supplies the protocol-frozen cross-setting evaluation, and AG News supplies prior task-specific evidence.

heterogeneous LoRA and cross-size knowledge transfer are not new. Selective-FD filters misleading client predictions and ambiguous ensembles (Shao, Wu, and Zhang 2024); FedID studies interactive distillation for pretrained language models under heterogeneous federated settings (Ma et al. 2023); FedHDS selects compact representative instruction data in federated LLM tuning (Qin et al. 2025); Fed-DFA prioritizes decision-boundary information (Wang et al. 2025); and heterogeneous-distillation temperature has been reconsidered explicitly (Qi et al. 2025). Most directly, FedFD shows that heterogeneous logit ensembles can be unstable and proposes aligned feature distillation (Li et al. 2025). We therefore do not claim that logit transfer failure is new. Our narrower contribution audits decoder-only LMs whose clients are also internally contaminated, separating training-example selection, canonical probability transfer, and server utility. Our severe-unfiltered arm is a FedMD/FedDF-style uniform canonical-probability ensemble-distillation control adapted to decoder-only verbalizer classes; it is not an exact reproduction of either algorithm.

Noisy labels and data selection. The early-learning behavior of deep networks motivates small-loss selection (Arpit et al. 2017). Co-Teaching exchanges peer-selected small-loss examples (Han et al. 2018); trusted clean data can also estimate noise corrections (Hendrycks et al. 2018). Superfiltering shows that a weak language model can select instruction data for a stronger model (Li et al. 2024). Delora decouples a dual-LoRA noisy-label detector from downstream PEFT (Yuan, Chen, and Zhang 2025), and FedClean performs clean-sample selection, multi-stage finite-class relabeling, and robust aggregation (Jiang and Zhang 2025); FedNed instead addresses extremely noisy clients by negative distillation (Lu et al. 2024). Targeted instruction selection such as LESS optimizes downstream influence rather than label purity (Xia et al. 2024). Thus purity–utility separation is not a general insight we originate. Our public-anchor and gradient scores are diagnostic interventions used to test whether higher audited purity predicts server or local utility in frozen heterogeneous-LM protocols; we do not claim to outperform these methods in their native settings.

Problem Setting

Clients $c \in \{1, \dots, C\}$ own decoder-only models f_c, tokenizers τ_c, and datasets D_c. The server model f_s may have a different architecture and tokenizer. A client dataset is a mixture

$$D_c = D_c^{\text{clean}} \cup D_c^{\text{conflict}} \cup D_c^{\text{off}},$$

where conflicting examples retain a target-task input but provide an incorrect answer, and off-task examples use a different task interface. Clients and server share a small public target-task anchor set A.

For architecture-agnostic classification exchange, let $\mathcal{Y}$ be the canonical label set and $L = |\mathcal{Y}|$. Client c transmits a probability vector over this label set for every anchor input. The client score vector contains token-normalized candidate-answer losses $\boldsymbol{\ell}_{ci}$, giving $q_{ci} = \text{softmax}(-\boldsymbol{\ell}_{ci})$ for anchor x_i. The server uses $\bar{q}_i = C^{-1} \sum_c q_{ci}$, computes its own canonical probabilities p_s, and minimizes $\mathcal{L}_s = \lambda_A \, \text{CE}(y_i, p_s) + \lambda_D \, \text{CE}(\bar{q}_i, p_s)$. The anchor-only control sets $\lambda_D = 0$; routed transfer uses the frozen λ_D. This interface is compatible by construction, but its utility depends on client data quality.

Training-Only Trust Endpoints

Class-conditional gradient routing

For a labeled example $z = (x, y)$, let $g(z)$ be the gradient of its candidate-answer loss with respect to local LoRA parameters and let a_y be the mean gradient of public anchors from

class y. Their agreement is

$$s_g(z) = \frac{g(z)^\top a_y}{\|g(z)\|\|a_y\|}.$$

Let $P(v_i) \in [0,1]$ denote the average-tie percentile rank of value v_i within client c, with lower values ranked first. For target-task examples, the inconsistency margin is $m(z) = \ell_y(z) - \min_k \ell_k(z)$; off-task examples receive one plus the largest observed target-task margin. The risk is

$$R(z) = \tfrac{1}{3}\{P(-s_g(z)) + P(\ell(z)) + P(m(z))\}.$$

Because raw cosine scales differ by model family, the routing gate uses the relative statistic

$$r_c = \frac{Q_{0.25}(\{s_g(z) : z \in D_c\})}{\max[Q_{0.25}(\{s_g(a) : a \in A\}), 10^{-12}]}.$$

The gate triggers when $r_c < 0.8$; otherwise all examples are retained. A triggered client keeps the lowest-risk $\max(1, \mathrm{round}(0.5|D_c|/L))$ examples per claimed target class, with source ID breaking ties; off-task examples have no claimed class and are excluded. Zero gradients are handled by the clamped cosine denominator. Thresholds and quotas were chosen on development data and fixed before the corresponding official-test runs. Clean-versus-conflict annotations are audit-only and never enter the score. Off-task membership is known from the explicit task interface; such examples have no claimed target class and are excluded whenever the gate triggers.

Evidence-Aware Experimental Design

Study	Design timing	Authorized role
TREC primary	Separately frozen, five seeds	Task-specific primary contrast
TREC four-arm	Post-primary, reused test	Discovery/mechanistic Holm family
IMDb four-arm	Frozen before official evaluation	Protocol-frozen primary contrast
IMDb matched-K	Post-development, pre-official	Secondary mechanistic Holm family
AG legacy	Three seeds plus post-observation check	Prior task-specific $D - A$ context
AG four-arm	Post-primary, run-disjoint rows	Secondary task-specific $D - C$ context

Table 1: Evidence ledger. Protocol chronology is supported by retained run records; hashes establish internal consistency, not external preregistration.

Table 1 fixes the authorized inferential role of each study; shared hashes or benchmarks do not upgrade discovery or descriptive evidence into protocol-frozen evaluation.

TREC discovery study. TREC-Coarse provides a six-way answer-type task (Li and Roth 2002); IMDb remains the off-task source. Exact duplicate training lines are removed by raw-line hash before a balanced allocation of 72 anchors, 72 development questions, and three disjoint 120-question client pools. Each client class contributes 8 clean, 8 cyclic-conflict, and 4 off-task slots. The complete 500-question official test is evaluated over five seeds. The task, manifest, runner, full arguments, canonical $[72, 6]$ probability payloads, and serialized wire payloads are content-addressed. The model families, two rounds, update budget, and routing thresholds match the AG News study. The parent primary protocol was frozen after instrumentation repairs and before its official evaluation; those repairs and invalidated smoke runs are recorded in its protocol history. The later four-arm protocol explicitly records prior test access and its own audit-counter repair before any retained full aggregate.

Protocol-frozen IMDb role reversal. IMDb sentiment becomes the binary target and AG News the off-task source. A content-addressed manifest separates 128 public anchors, 1,000 development reviews, three disjoint 256-example client pools, and the complete 25,000-review official test. Each severe client contains 103 clean, 102 cyclic-conflict, and 51 off-task examples. Three development seeds authorize implementation only; no accuracy direction may tune or stop the complete ten-seed official matrix. The single primary contrast is D–C accuracy, using a two-sided paired Student-t test and 95% seed interval. The narrow negative-direction gate contains exactly three frozen clauses: routing must raise the audited clean fraction above 103/256, the paired Student-t interval for D–C must be strictly negative, and the two-sided paired p-value must be below .05. The test was not decoded for model assessment until the protocol, development matrix, verifier, and analysis rule had been frozen.

After development but before official-test access, we froze a separate matched-size secondary addendum. Random-K, routed-K, and maximum-purity-K all select $K = 128$ with 64 examples per claimed label and share the severe mixture, optimizer steps, and server settings. Maximum-purity-K contains all 103 clean examples plus 25 conflicts and is therefore not an oracle-clean arm. Routed-minus-random and maximum-purity-minus-routed accuracy form exactly one two-sided Holm family; they cannot alter or confirm the original primary.

AG News task-specific context. AG News is the four-way target task and IMDb supplies real off-task text (Zhang, Zhao, and LeCun 2015; Maas et al. 2011). Conflicts use a deterministic non-identity label permutation. A frozen manifest separates 256 public anchors, 800 development examples, disjoint client pools, and the full 7,600-example official test. Clients are Qwen2.5-0.5B, SmolLM2-360M, and TinyLlama-1.1B; the server is Qwen2.5-1.5B. Its original three-seed comparison and five-seed same-test direction check precede a post-primary four-arm train-derived secondary study; each retains its own evidence label.

Controls and statistics. Within each server task, comparisons match server initialization, public anchors, server optimizer steps, prompts, and seed. The original AG News primary and seed replication execute shadow unfiltered clients but set $\lambda_D = 0$, so their recorded probability payload is unnecessary to the server rather than absent. The TREC anchor

and the later AG four-arm runner omit the client path. Cost ratios are therefore protocol-specific and do not share one end-to-end work definition. Each run-level protocol is serialized before its corresponding execution; this does not make later phases globally preregistered once an earlier phase has accessed the same benchmark test. Paired Student-t intervals characterize seed variation, not task- or sample-level uncertainty. TREC uses five designated primary seeds; its disclosed reused-test decomposition uses ten new common seeds and three adjacent two-sided accuracy contrasts in one Holm family. IMDb uses ten common seeds, one original primary contrast, and the separate two-contrast secondary Holm family above. All other server metrics and net contrasts are descriptive. Raw records carry evidence labels, model and manifest hashes, split scope, wall time, and peak GPU allocation.

Metrics and optimization. For TREC, IMDb, and the new AG four-arm runner, class scores are negative token-normalized answer-suffix losses; we report accuracy, macro-F1, the true-label suffix loss, and 10-bin ECE from the resulting canonical scores. Because token-normalized suffix losses are renormalized across verbalizers, this is canonical verbalizer-score ECE, not calibration of the LM's native sequence probability. We additionally report normalized canonical class NLL where the runner records it. The legacy AG endpoint named `label_nll` instead records target-answer suffix loss and is not a canonical class NLL. All adapters target query/value projections with LoRA rank 4, scale 8, and no dropout, optimized by AdamW. The AG holdout uses two rounds of 40 client and 40 server steps (batch 8; server learning rate 10^{-4}); IMDb matches this budget. Frozen JSON protocols contain the remaining length and selection settings. Peripheral local-adaptation protocols are detailed in the supplementary material.

Results

TREC discovery: a net comparison hides opposing effects

Across five primary seeds, routed-minus-anchor accuracy ranges from -1.8 to $+2.6\,pp$ with mixed signs. The mean is $+0.40\,pp$, with descriptive 95% seed interval $[-1.74, +2.54]\,pp$ and two-sided paired $p = 0.631$. Macro-F1 changes by $+0.73\,pp$ with an interval that also crosses zero. Canonical class NLL decreases by 0.0304 ($p = 0.0616$), while the secondary score ECE decreases by 0.0373 in every seed, with interval $[-0.0487, -0.0259]$ and nominal, multiplicity-unadjusted $p = 0.00081$. Thus TREC does not validate an accuracy benefit; the interval spans modest harm and benefit. Its lower score ECE is a descriptive secondary pattern. Routing triggers in 27/30 client-rounds and records 19,830 bytes of serialized probability tensors per run, excluding transport metadata. Its $9.61\times$ mean timed-region per-process wall-time ratio was measured with anchor runs serial and routed runs up to two-way concurrent; it is neither end-to-end nor an equal-work deployment benchmark.

After observing this compound comparison, we froze a post-primary four-arm ablation on the already accessed TREC test with ten new common seeds. The three adjacent accuracy contrasts form one two-sided Holm family. Clean uniform transfer versus anchor-only is inconclusive ($-0.20\,pp$, unadjusted interval $[-0.71, +0.31]$, Holm $p = .397$). Contrary to a simple contamination-harm story, severe unfiltered transfer exceeds clean transfer by $1.58\,pp$ (unadjusted $[+0.54, +2.62]$, Holm $p = .015$). The complete routed intervention then reduces severe-uniform accuracy by $1.42\,pp$ (unadjusted $[-2.19, -0.65]$, Holm $p = .0074$), while descriptively reducing class NLL by 0.0314 and score ECE by 0.0596. The routed-minus-anchor net is $-0.04\,pp$ with interval $[-0.84, +0.76]$, without a fourth test. This same-test analysis diagnoses an accuracy–canonical-score trade-off; it is not a new holdout or independent validation and does not isolate which mixture component produces the severe-uniform gain. Routing triggers in 57/60 client-rounds and raises the selected clean fraction from 40.0% to 64.6% when off-task rows remain in the denominator, yet lowers accuracy relative to severe uniform transfer. Because this decomposition was designed after access to the same 500-example test, it motivated the negative directional prediction evaluated next rather than independently confirming it.

Protocol-frozen IMDb primary analysis

The complete frozen ten-seed matrix gives mean accuracies of 91.83%, 91.79%, 91.10%, and 91.39% for arms A–D, respectively. The pre-specified narrow cross-task gate **fails**: the frozen severe-mixture baseline is 40.23% clean, the observed severe-uniform share is 40.23%, and the routed share is 66.53%. The primary D–C effect is 0.29 pp (95% CI [0.07, 0.51]; paired two-sided t-test $p = 0.0149$; exact sign-flip sensitivity $p = 0.0176$). The strictly positive D–C interval contradicts the TREC-derived negative directional prediction in this protocol-frozen IMDb evaluation; audited purity and server accuracy move in the same direction in this setting.

Matched-K secondary mechanistic analysis

This separately frozen secondary family holds $K = 128$, claimed-label quotas, local steps, and the severe mixture fixed. Audited clean shares are 49.66% for random-K, 71.02% for routed-K, and 80.47% for maximum-purity-K. The routed-K minus random-K effect is 0.32 pp (95% CI [0.17, 0.48]; raw $p = 0.0010$, Holm $p = 0.0020$). The maximum-purity-K minus routed-K effect is 0.05 pp (95% CI [-0.10, 0.19]; raw $p = 0.4872$, Holm $p = 0.4872$). After Holm correction, routed-K significantly outperforms random-K. Maximum-purity-K does not show a statistically detectable gain over routed-K. These data therefore do not establish an additional accuracy gain from the higher audited clean fraction. These are secondary mechanistic tests and do not confirm or replace the primary analysis.

AG News task-specific context

Across the original three seeds, routing loses $-0.56\,pp$ relative to the anchor-only server. The descriptive interval runs from -0.76 to $-0.36\,pp$, and every paired difference is negative. Routed distillation lowers answer loss and score ECE but costs $2.01\times$ the matched wall time. Accuracy and macro-F1 both decline; adding routed client targets was therefore

Arm	Accuracy (%)	Macro-F1 (%)	Score NLL	Score ECE
Anchor only	91.83 ± 0.10	91.83 ± 0.10	0.2470 ± 0.0088	0.0436 ± 0.0022
Clean uniform	91.79 ± 0.11	91.78 ± 0.12	0.2099 ± 0.0025	0.0193 ± 0.0039
Severe uniform	91.10 ± 0.18	91.10 ± 0.18	0.2578 ± 0.0043	0.0707 ± 0.0053
Severe routed	91.39 ± 0.26	91.39 ± 0.26	0.2139 ± 0.0068	0.0115 ± 0.0028
Random-K	91.14 ± 0.13	91.13 ± 0.13	0.2558 ± 0.0063	0.0690 ± 0.0067
Routed-K	91.46 ± 0.23	91.46 ± 0.23	0.2118 ± 0.0061	0.0101 ± 0.0023
Maximum-purity-K	91.51 ± 0.16	91.51 ± 0.17	0.2122 ± 0.0041	0.0125 ± 0.0040

Frozen contrast	Mean pp [95% CI]	Raw p	Holm p	Exact p
Routed − severe uniform (primary)	+0.29 [+0.07, +0.51]	0.0149	—	0.0176
Routed-K − random-K (secondary)	+0.32 [+0.17, +0.48]	0.0010	0.0020	—
Maximum-purity-K − routed-K (secondary)	+0.05 [-0.10, +0.19]	0.4872	0.4872	—

Table 2: Complete official IMDb results over ten common seeds. Values in the arm panel are mean ± sample SD. Confidence intervals are unadjusted paired two-sided 95% Student-t intervals.

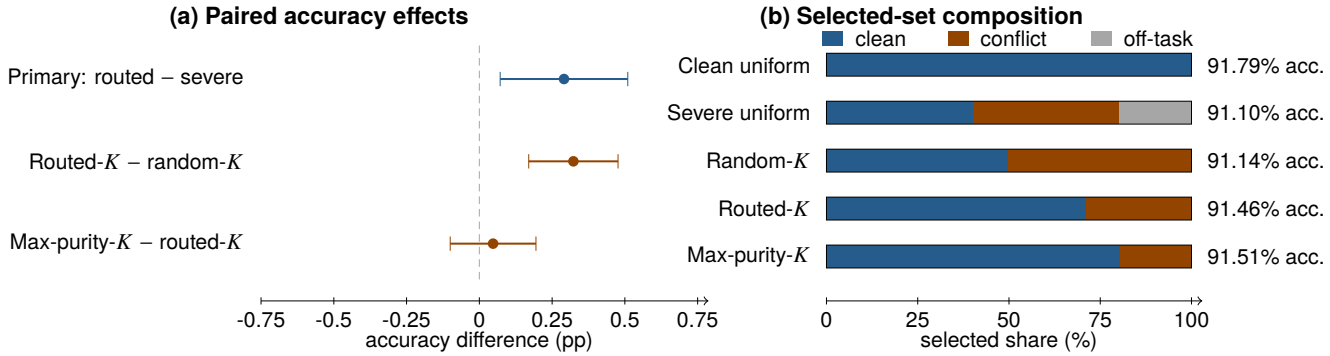


Figure 2: Paired IMDb effects and audited selected-set composition. Points are mean paired accuracy differences and bars are unadjusted 95% intervals; the two matched-K tests form the frozen secondary Holm family.

lower under AG News accuracy in this pipeline, not uniformly worse across predictive metrics. After observing this result, we froze five additional seeds without retuning. All five differences are negative, with mean $-0.80\,pp$; its descriptive interval runs from -1.18 to $-0.42\,pp$. The predeclared 4-of-5 directional heuristic passes, but it is not an alpha-controlled test and reuses the accessed official test. The combined eight-seed estimate ($-0.71\,pp$) is descriptive.

The post-primary AG News train-derived secondary holdout gives the same four-arm decomposition on 4,000 examples that are disjoint from this runner's anchors, development set, and clients, but were selected after the primary result; 303 examples had appeared in at least one earlier server-client allocation. It is therefore run-disjoint same-task resampling, not external validation. Over ten seeds, B–A is $+0.01\,pp$ (95% CI $[-0.28, +0.29]$, Holm $p = .954$), C–B is $-2.04\,pp$ ($[-2.44, -1.65]$, Holm $p < 10^{-4}$), and D–C is $+1.46\,pp$ ($[+1.06, +1.86]$, Holm $p < 10^{-4}$). The descriptive D–A net remains $-0.58\,pp$ ($[-0.80, -0.35]$). AG therefore matches IMDb, not TREC, on the positive D–C sign while preserving the older AG endpoint harm relative to anchor-only.

At development seed 42, severe AG News mixtures reduce local accuracy by 21.75–25.88 points across the three client families. Across Qwen seeds 42–44, routing improves severe local accuracy by $+3.37$ to $+30.00\,pp$, but this local improvement does not translate to the server endpoint. The detailed compatible-model and local-adaptation controls are reported in the supplementary material; they are not additional federated server endpoints.

Discussion

Architecture compatibility and transfer utility answer different questions. Canonical probabilities create a common interface across model families, but their effect still depends on the target task, contamination, selection rule, and evaluation metric. The TREC decomposition illustrates why a routed-minus- anchor endpoint is insufficient: a near-zero net contrast can conceal opposing clean-transfer, contamination, and routing effects. Its lower descriptive score NLL and score ECE also cannot be substituted for accuracy utility.

The four-arm audit separates these intervention-level contrasts. B–A asks whether clean client probabilities help beyond the public anchor; C–B asks what changes under the joint severe mixture; and D–C measures the complete routing intervention. Most importantly, the protocol-frozen IMDb result reverses the TREC-derived D–C direction: both tasks show large purity enrichment, but routing reduces accuracy

on the reused TREC test and improves it on IMDb. The frozen harm hypothesis therefore fails rather than replicates. This is evidence of setting-specific heterogeneity: the TREC direction did not transport to IMDb, but the two settings do not identify a general task-level law.

At matched size, routed-K improves over target-eligible random-K, demonstrating useful ranking signal on IMDb. Yet the maximum-purity control raises clean share further without a detectable gain over routed-K. Consequently, neither "more pure always helps" nor "purity is unrelated to utility" is licensed. Purity alone is insufficient to determine the sign or magnitude of server utility. Component-wise attribution remains unresolved because the router combines gradient agreement, loss, and inconsistency ranks.

Inferential role is part of the result. TREC's reused-test decomposition is hypothesis-generating even with multiplicity control; the protocol-frozen IMDb role reversal supplies the cross-setting stress test; the legacy AG News comparison is prior task-specific evidence with a different $D - A$ estimand; and the completed AG four-arm holdout is secondary post-primary same-task resampling. Hashes and complete-matrix checks establish internal consistency, not external preregistration. The resulting claim is therefore an evaluation diagnosis rather than a new federated optimizer or a universal law relating purity and utility.

Limitations and Ethics

The study simulates three clients from a fixed decoder-only model suite; it does not sample organizations, architectures, or tasks from a population. Conflict and off-task rows are controlled interventions, and the severe arm changes both simultaneously, so C–B cannot attribute an effect to either component. Seed intervals describe optimization variation rather than example- or task-population uncertainty. TREC's four-arm result reuses an accessed 500-example test, the IMDb matched-size family was designed after development, and AG's secondary rows are not an independent validation. The matched routed arm forces $K = 128$, whereas the original router falls back to all 256 examples when its gate does not trigger; four of ten official seed trajectories therefore diverge exactly as allowed by the frozen conditional contract. A post-record verifier repair restored that conditional check after the earlier verifier incorrectly required unconditional full trajectory identity; no protocol, runner, raw record, seed, or statistic was changed. Canonical verbalizer-score ECE is not native sequence-probability calibration. Pretraining contamination cannot be ruled out, and TREC's published license is recorded as unknown. Finally, logits, losses, anchors, and hashes provide no formal privacy; membership or reconstruction leakage is not measured. We make no privacy-preserving, production, or broad-generalization claim.

AI-assistance disclosure. OpenAI Codex assisted with code drafting, experiment orchestration, artifact validation, and manuscript editing. The human authors are responsible for the study design and submission, verified the cited sources and reported results against the raw artifacts, and take full responsibility for the paper's content.

Conclusion

Architecture-compatible probability exchange solves an interface problem, not the problem of deciding which client supervision benefits a server. Our four-arm audit exposes opposing effects that a routed-minus-anchor endpoint can hide. Routing enriches clean examples on both TREC and IMDb, yet its accuracy effect changes sign across the evaluated configurations; on IMDb, score-based routing beats matched random selection, while the higher audit-only maximum-purity control does not show a statistically detectable additional gain. Selection purity is therefore an important composition diagnostic but not a sufficient proxy for transfer utility. Future heterogeneous adaptation work should report composition, canonical score quality, and downstream utility separately under complete, evidence-labelled intervention matrices.

Reproducibility

The artifact contains content-addressed manifests and protocols, model-weight hashes, deterministic mixtures, per-run raw JSON, generated summaries, and reconstruction tests. Records are accepted only after complete-matrix verification of identity, data splits, arguments, protocol and runner digests, canonical probability payloads, selection composition, and evidence labels. Paper numbers and figures are regenerated from verified records rather than copied from console logs. The TREC primary and reused-test matrices, IMDb original and matched-size matrices, and their distinct statistical families remain separate. Detailed configurations, invalidated-run history, software versions, and cost diagnostics accompany the artifact and supplementary material.

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